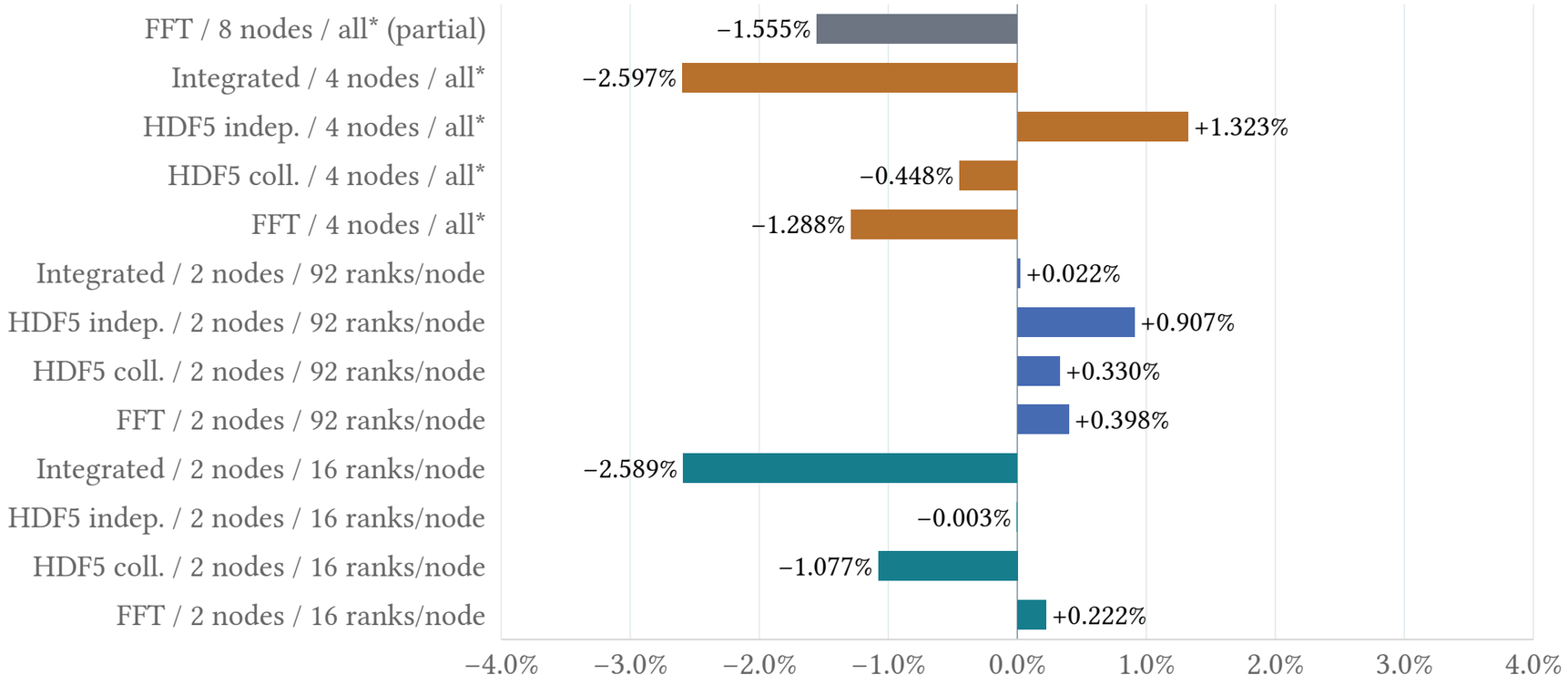
# Compiler hardening: results so far

## Completed point estimates: −2.60% to +1.32%

Full = candidate hardening profile. Reference = ordinary build. Negative = faster. Positive = slower.




### Coverage

12 complete conditions
1 partial
3 not measured

Missing: 8-node I/O cases

### Sample counts

10 pairs per case
(20 process runs)

Partial: 4/10 pairs

One pair = full + reference

Provisional screenshot transcription, 16 Oct 2026. Error bars unavailable. Original build verification pending.

Takeaway: no consistent slowdown is apparent in completed cases. Methods and next tests follow.

Small point estimates do not prove zero overhead. *Exact 4-/8-node core counts are not visible in the photos.

# Test design and measurement

Dense MPI phase only. Synthetic library workflows. Earlier serial/package screens are separate.
Full and reference change the caller plus the complete trial-owned library stack.

## Workloads constructed

| | |
|---|---|
| **FFT loop** | 3D forward/inverse complex FFTs. Plans reused. |
| **HDF5 collective** | Collective writes, flush/close and read-back. |
| **HDF5 independent** | Independent writes, flush/close and read-back. |
| **Integrated loop** | Repeated FFTs with periodic checkpoints. |

## How the comparison works

Maximum-rank loop time excludes startup, planning and final read-back.
Balanced process pairs keep work and rank layout fixed within each pair.
One rank per physical core. Slurm/Open MPI uses the saved UCX fabric policy.

Exact sizes, runtime settings and build treatment await original-record verification.

# Time interpretation and next tests

Small point estimates warrant further checks before a deployment decision.

## Percentages and elapsed time

Change = 100 x (full/reference - 1).
Positive: slower. Negative: faster.

## Example only: a 12-hour reference job

At +1%: 7 min 12 sec longer.
At −1%: 7 min 12 sec shorter.

This assumes the same workload mix and a sustained effect across the whole job.
Longer jobs accumulate more seconds.
The percentage stays the same.

The photos do not establish actual test duration.
These time examples are conditional arithmetic.

## Follow-up coverage

### Build verification and deployment comparison

Audit flags and raw paired records. Use one fixed ordinary caller with each library stack at 20 nodes.

### FFT grids and sustained work

Vary transform sizes and loop duration. Check whether any effect persists as work increases.

### I/O patterns

Vary checkpoint size, cadence and read/write mix. Cold-cache and durable-storage cost remain open.

### MPI and repeatability

Add OSU. Repeat in independent allocations. Whole-application impact still needs evidence.